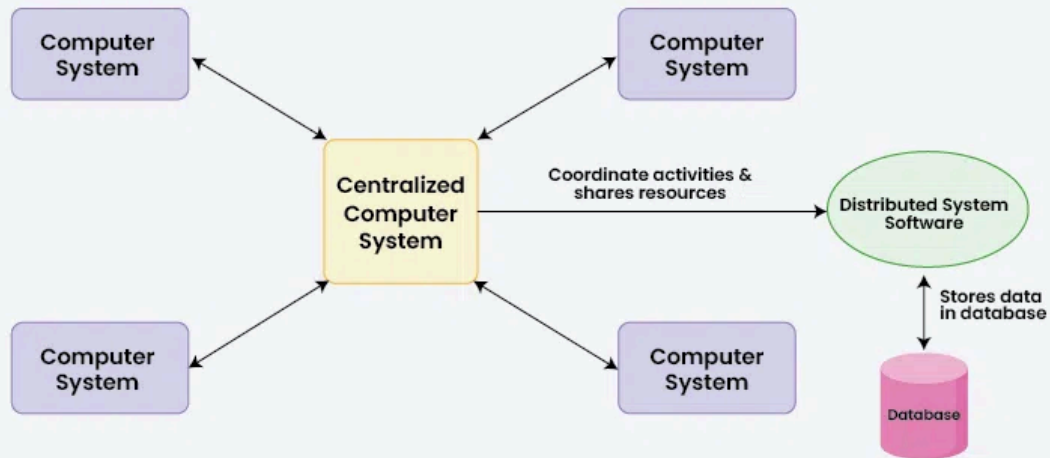
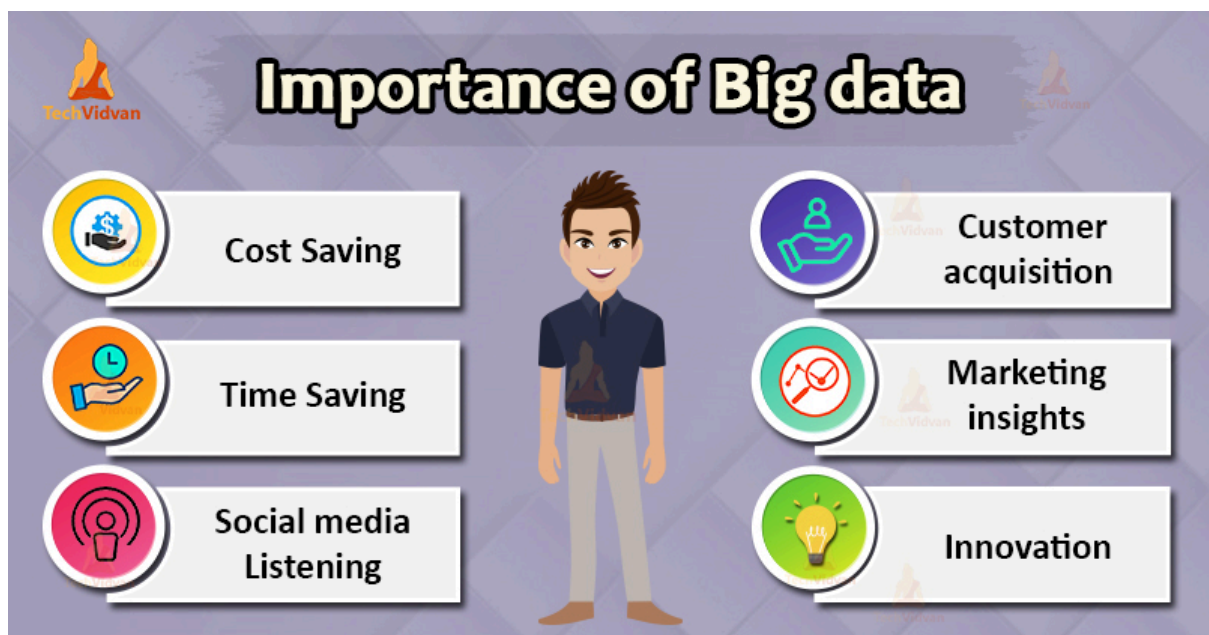


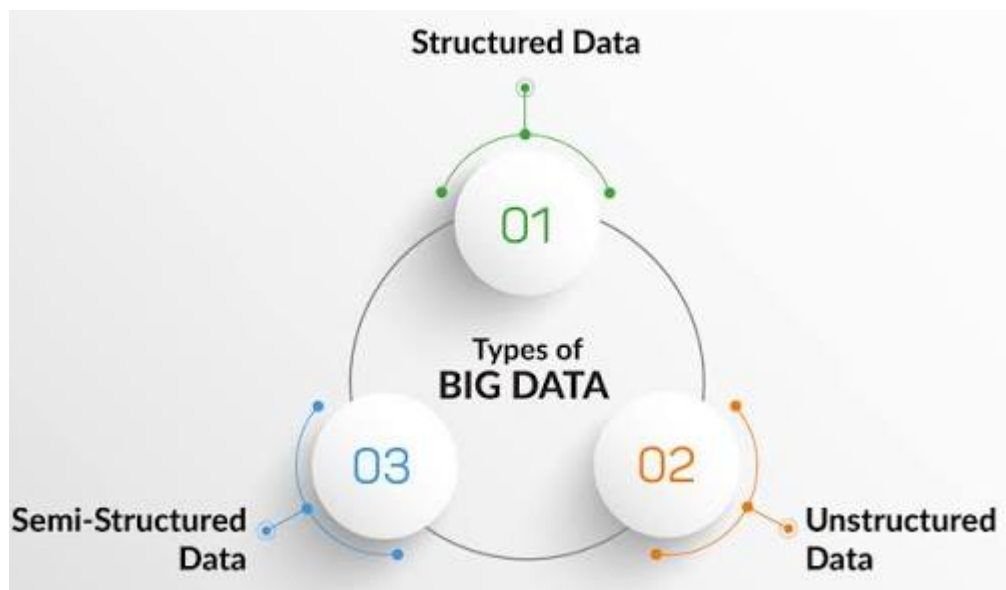
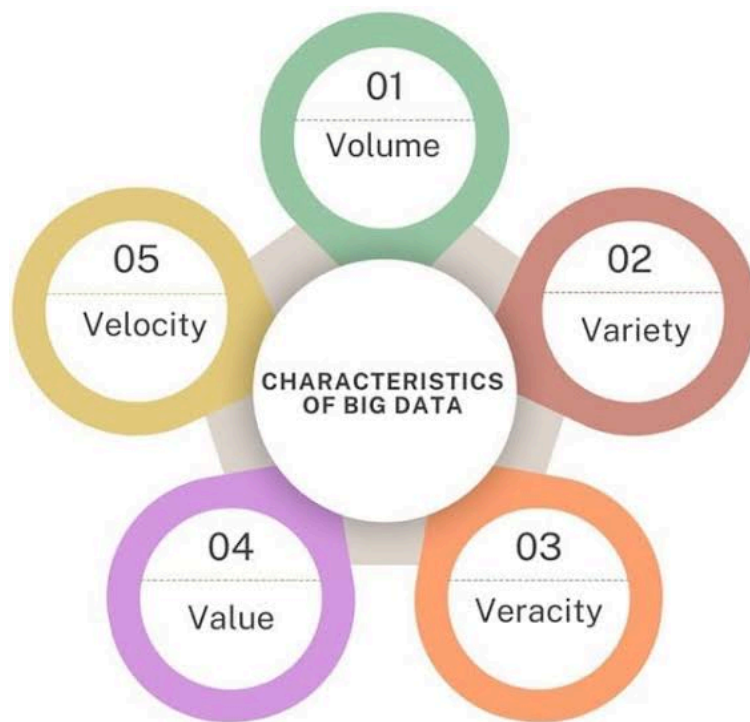
What is a Distributed File System?



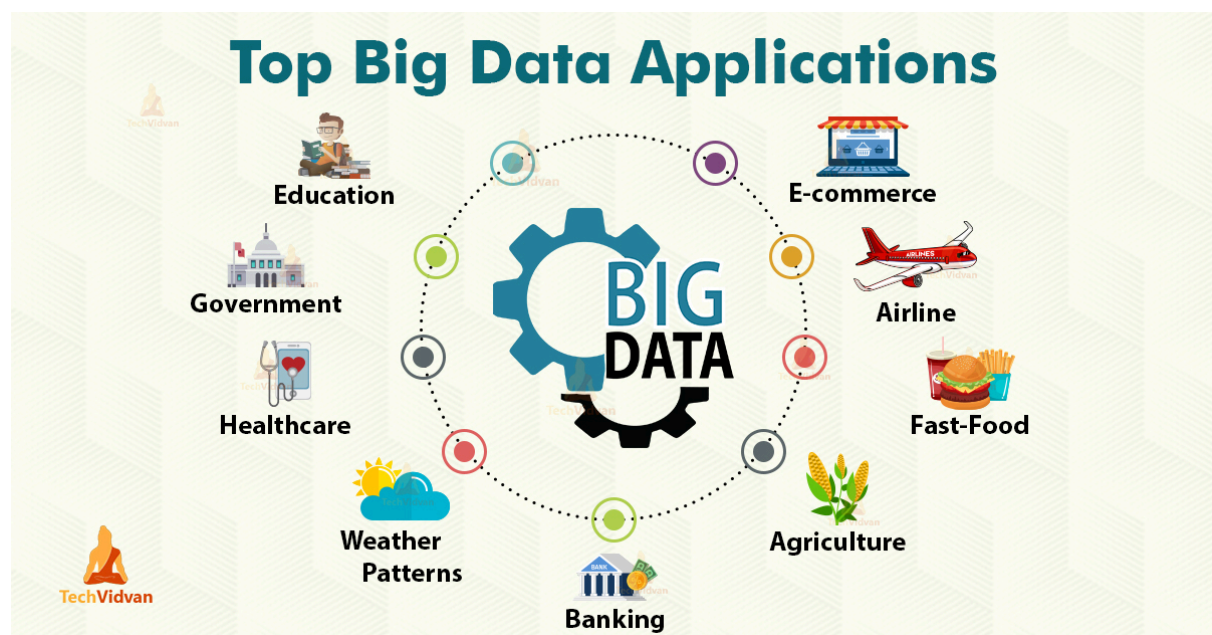
- Design Issues of Distributed System

- Scalability
- Reliability
- Availability
- Consistency
- Latency
- Load Balancing
- Security
- Architectural Design Patterns
- Communication Issues
- Data Management

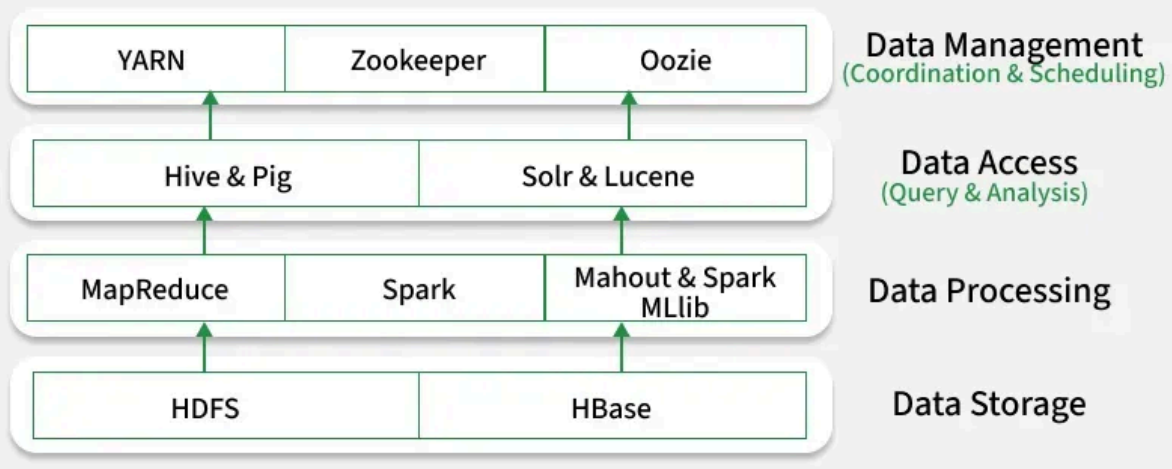




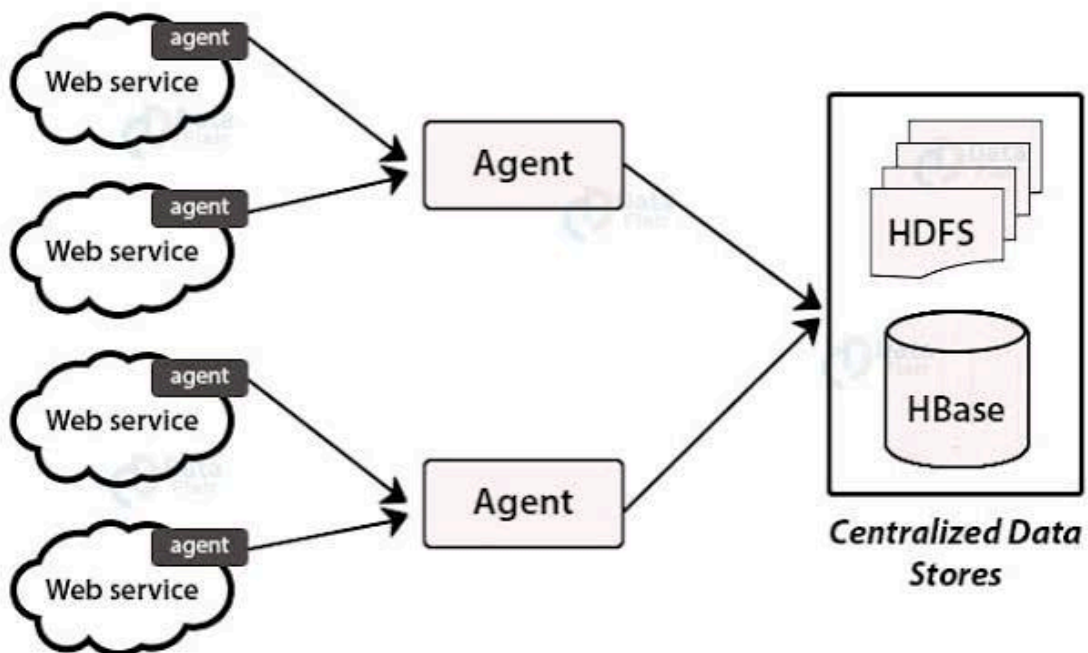
Components	Big data	Traditional Data
Queries	Largely Abandoned SQL	Traditional SQL
Architecture	Distributed	Centralized
Data Types	Structured, Semi- Structured and Unstructured	Structured
Data Model	No Schema	Fixed Schema
Data Relationship	Unknown or complex	Known Relationship
Data volume	Petabytes or Exabytes	Terabytes
Data Traffic	More	Less
Data Integrity	Less	High

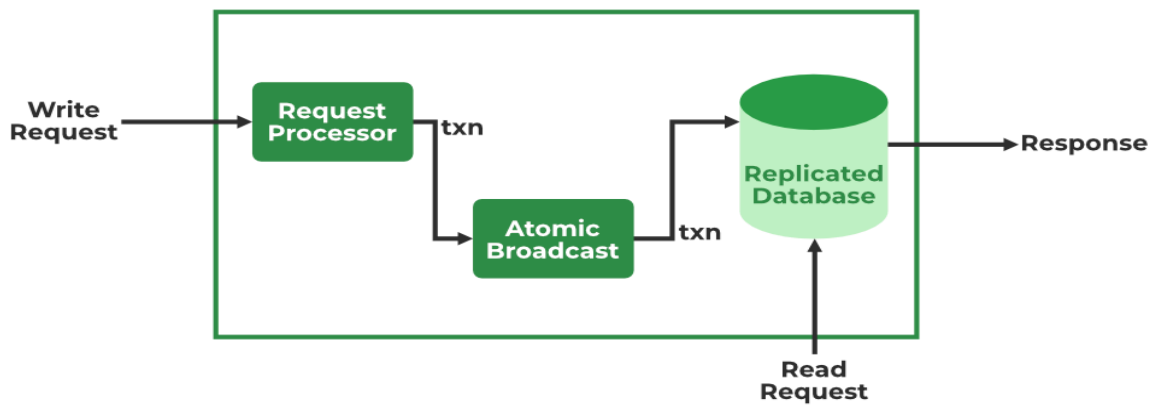
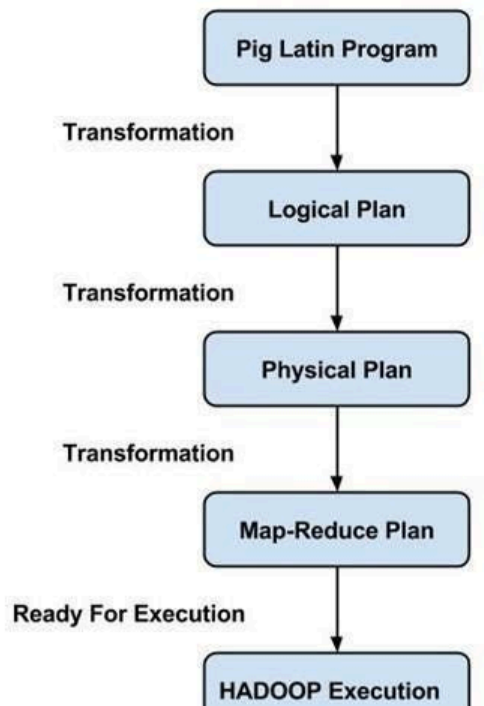


Hadoop Ecosystem

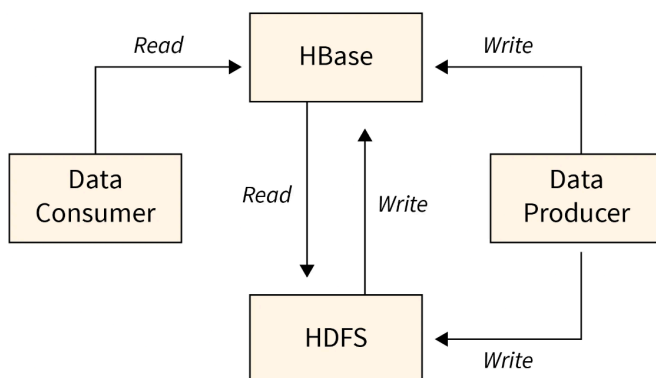


Apache Flume Data Flow

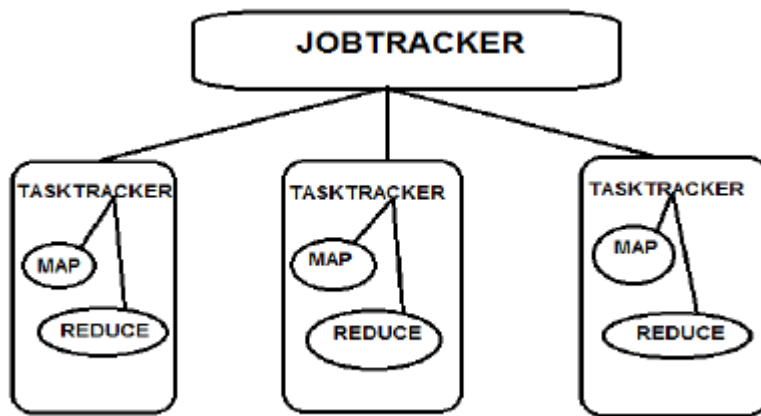
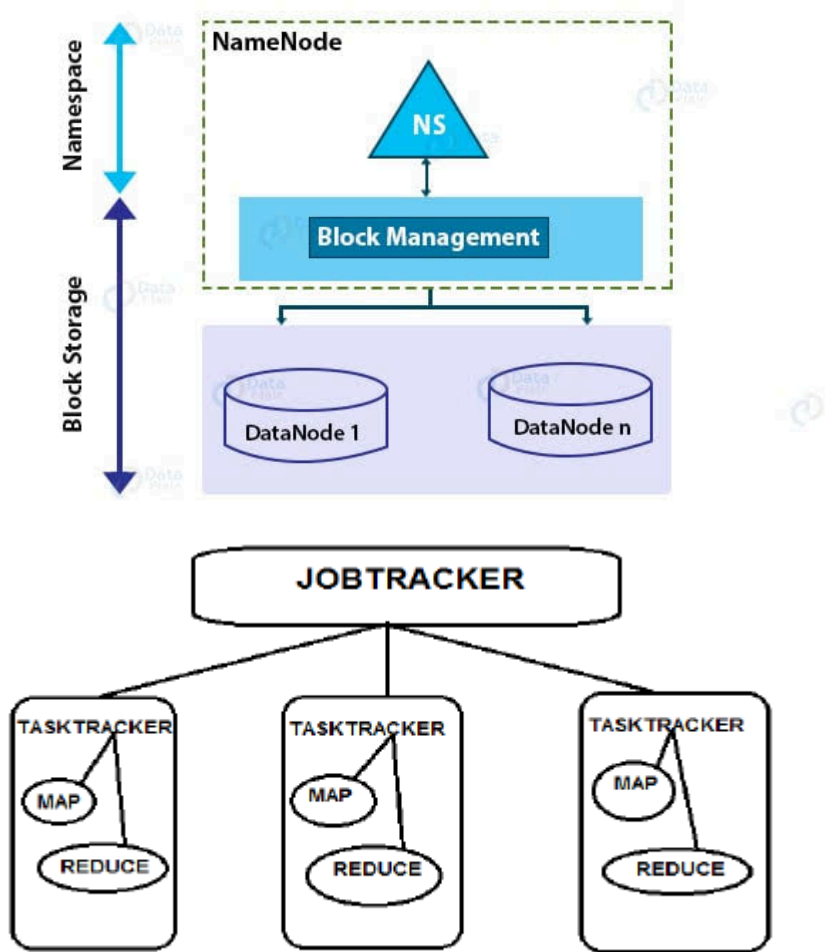




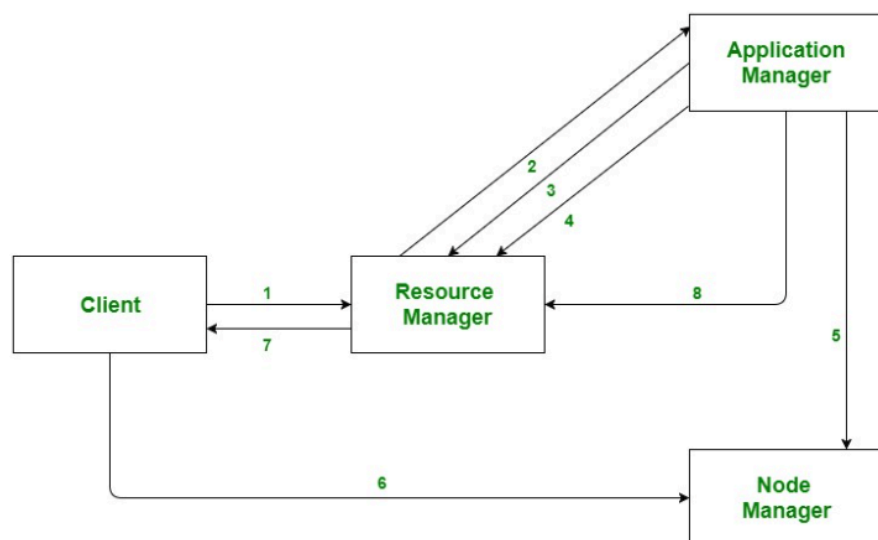
Zookeeper uper wla diagram

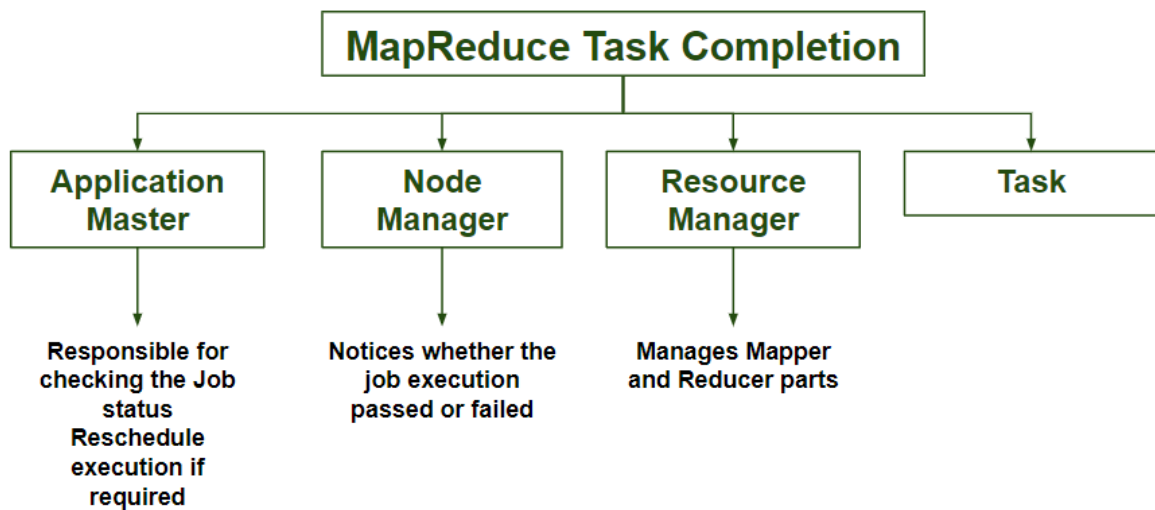
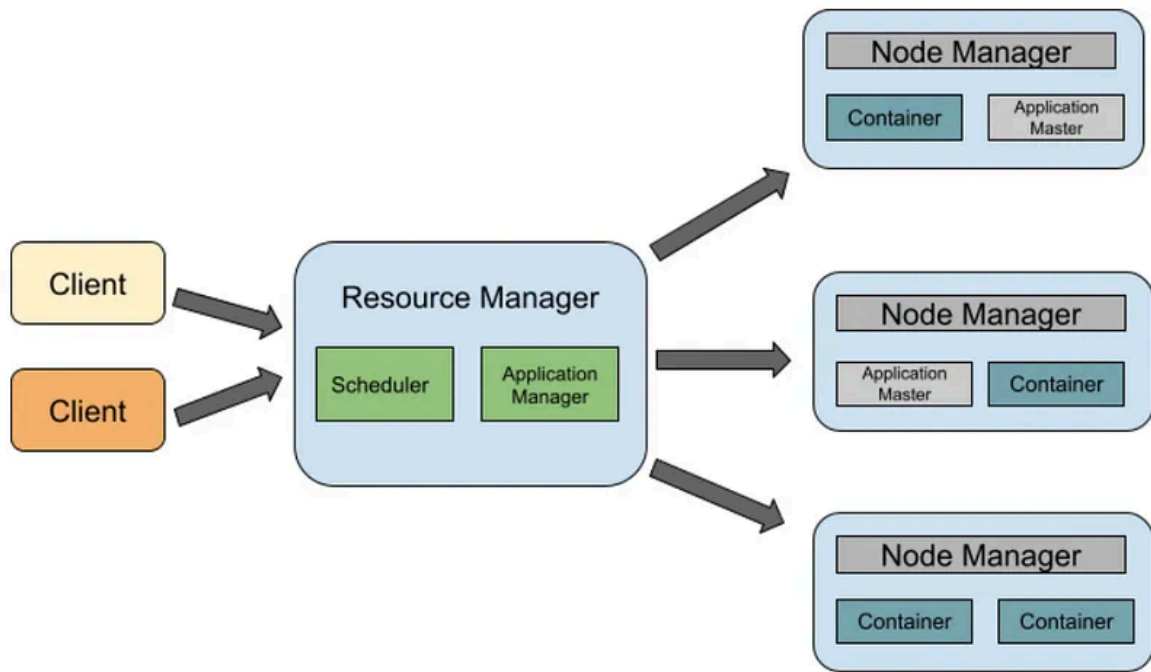


HDFS Layers



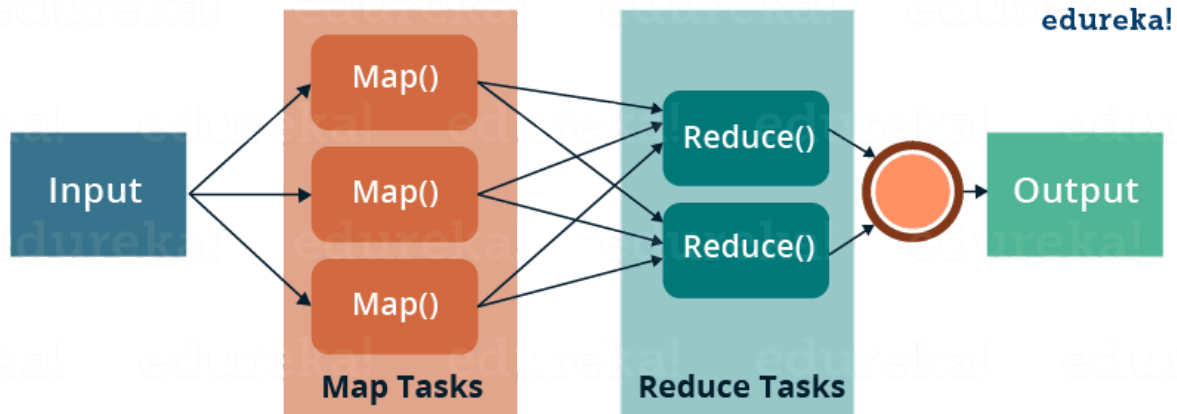
Application workflow in Hadoop YARN:



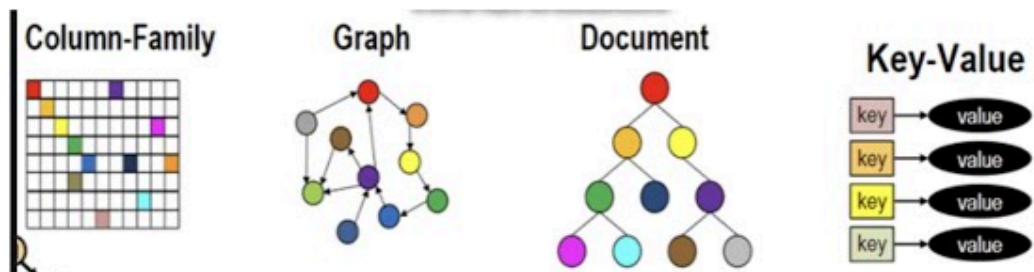


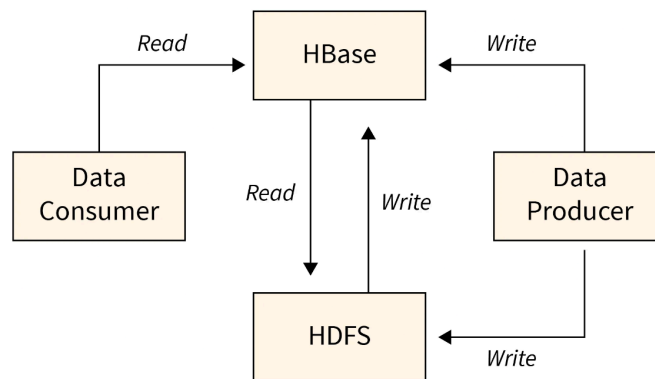
$$\begin{bmatrix} A & B \\ C & D \\ E & F \end{bmatrix} \times \begin{bmatrix} G \\ H \end{bmatrix} = \begin{bmatrix} A \times G + B \times H \\ C \times G + D \times H \\ E \times G + F \times H \end{bmatrix}$$

The diagram shows the matrix multiplication of a 3x2 matrix by a 2x1 matrix. Blue arrows highlight the calculation of the first row: $A \times G$ and $B \times H$. An orange arrow highlights the calculation of the third row: $E \times G$ and $F \times H$.



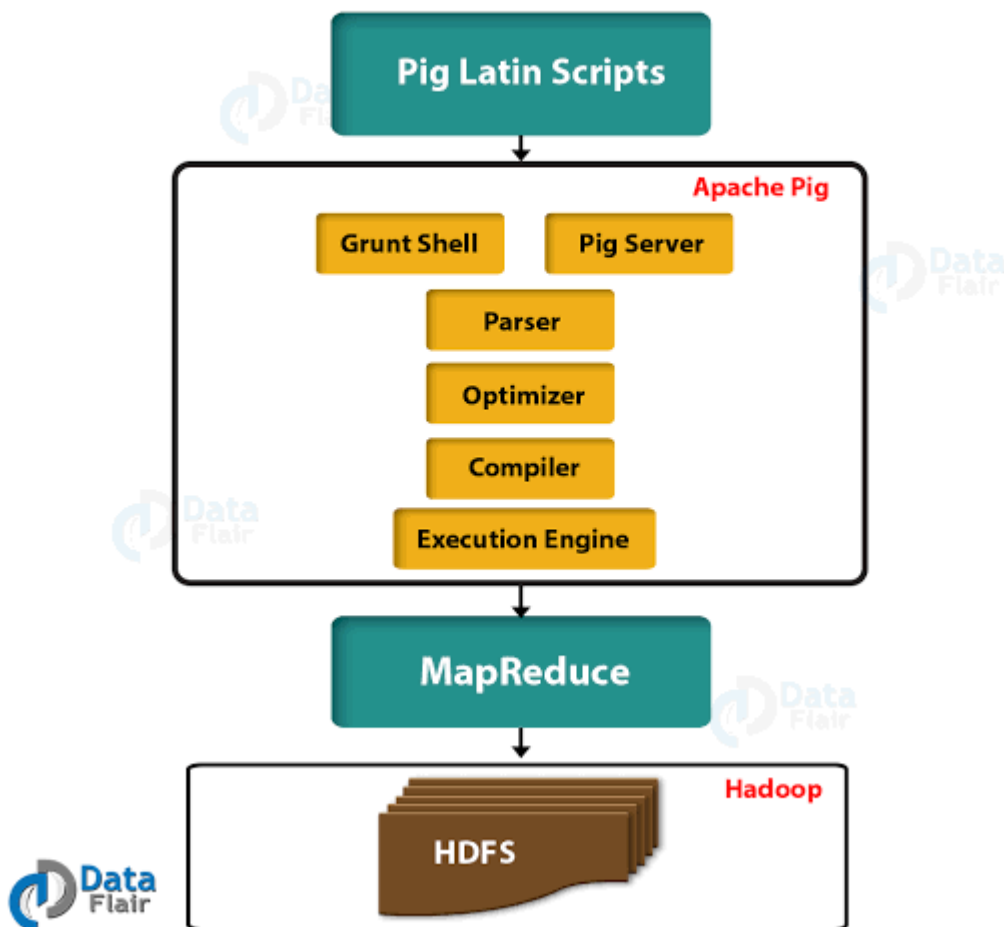
NoSQL





SCALER
Topics

Architecture of Apache Pig



The ETL Process Explained

